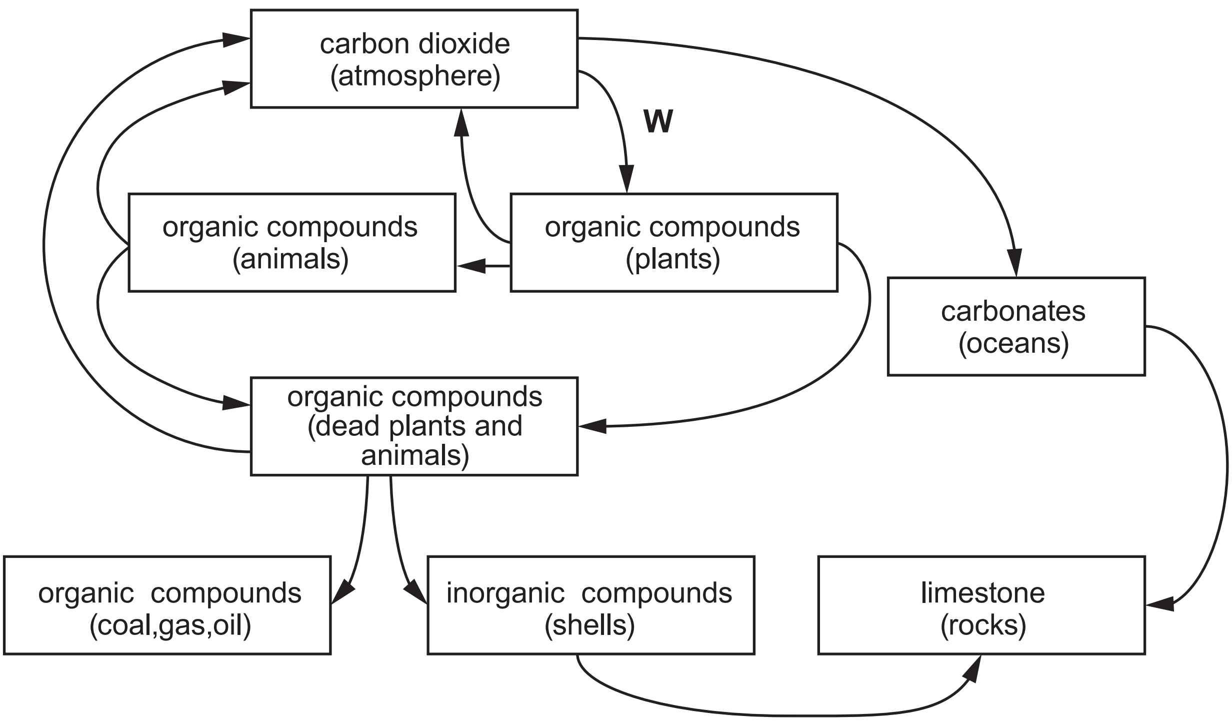
Examiner only

6. **Image 6.1** shows a natural carbon cycle unaffected by human activity. The arrows represent processes which transfer carbon from one form to another.

Image 6.1



(a) (i) Process **W** involves the fixation of carbon in green plants. Name the **two** reactants involved and the enzyme that catalyses this process. [2]

Reactants ....

....

Enzyme ....

(ii) I. On **Image 6.1, label one of the arrows** with an **X** to show a process which involves micro-organisms.

II. Explain the role of micro-organisms in this process. [2]

....

....

....

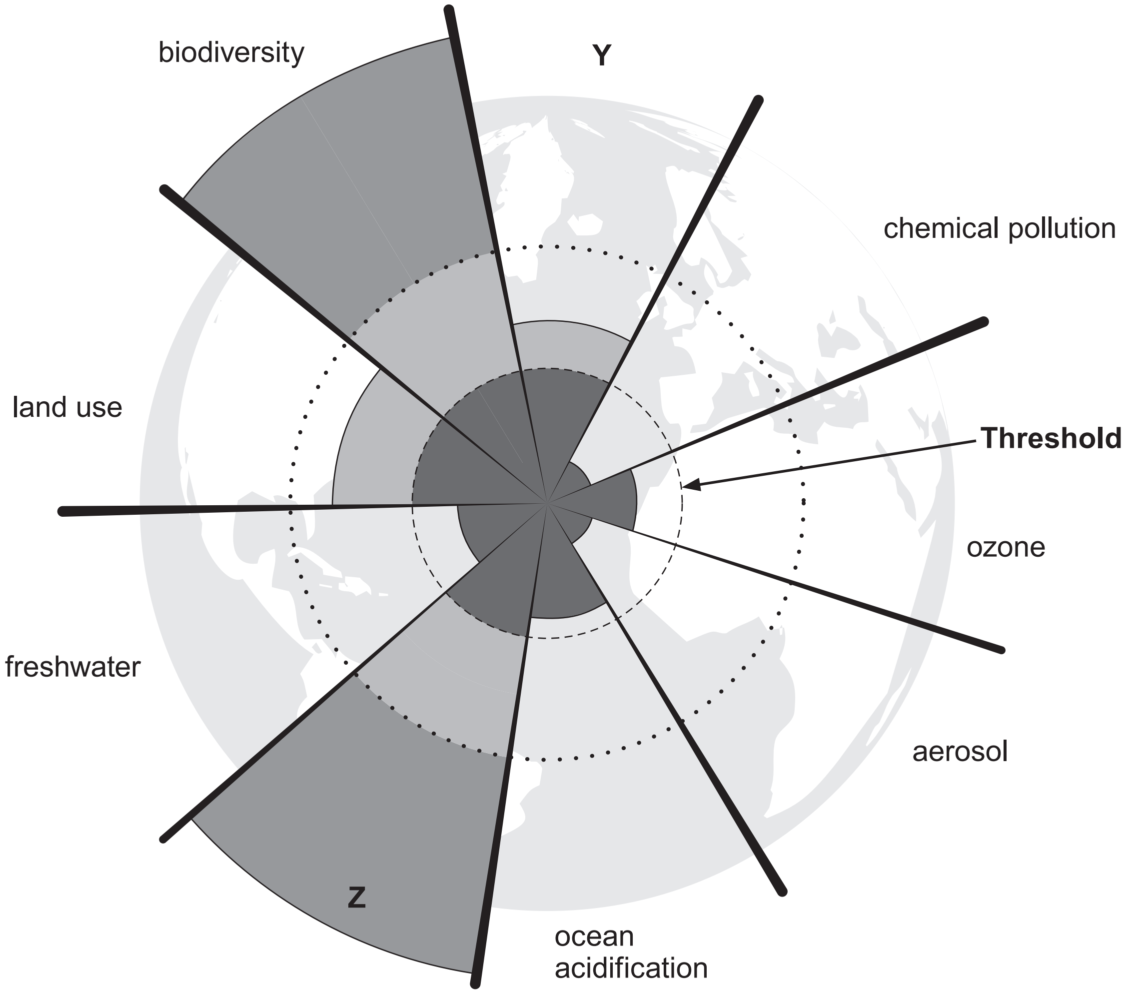
(iii) **Add an arrow labelled C** to **Image 6.1** to represent the transfer of carbon as a result of human activity. [1]

(b) Nine global systems have been identified as being key regulators of the Earth's stability. Values have been proposed that represent boundaries or thresholds. **Table 6.2** shows two of the nine systems together with their threshold values and current values and **Image 6.3** displays the threshold values and current values as a circular graph.

Table 6.2 – Planetary Boundaries

Planetary System	Parameters	Threshold values	Current value
Climate change	Atmospheric carbon dioxide concentration (ppm by volume)	350	387
Nitrogen	How much nitrogen is removed from the atmosphere for human use (tonnes $\times 10^6$/year)	35	121

Image 6.3



Examiner only

(i) Use the information in **Table 6.2** to name the **two** missing planetary systems labelled **Y** and **Z** in **Image 6.3**. [1]

Y .... **Z** ....

(ii) Use **Image 6.3** to state what the **two** planetary systems in **Table 6.2** have in common with each other and with the **Land-use** system. [1]

....

....

(c) Explain what is meant by a safe operating space for humanity, describe where that is shown in **Image 6.3**, and describe the consequences of exceeding planetary boundaries. [3]

....

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